

Scalability & Future Plan

Date	01 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the proposed solution for **A Comprehensive Measure of Well-Being** can be scaled to support larger datasets, increased user traffic, and additional analytical capabilities in the future. It also presents the team's roadmap for enhancing and evolving the project beyond its current implementation, ensuring long-term usability, maintainability, and impact..

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The system currently relies on a single HDI dataset and a basic Linear Regression model.	Prediction accuracy may decrease when handling diverse development patterns across countries.	High
2	The application provides only basic HDI prediction functionality without advanced analytics and visual reporting.	Users receive limited insights into development trends and contributing factors.	Medium

3	The current Flask application is designed primarily for academic and small-scale usage.	Performance may reduce when handling a large number of concurrent users.	Low
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Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Application supports limited concurrent users	Deploy on cloud platforms (Render/AWS) with autoscaling to handle more users
2	Data Storage	HDI data is stored locally in CSV format.	Integrate PostgreSQL or MySQL databases for improved scalability and reliability
3	Performance	ML prediction and data processing can be slow with large datasets	Optimize code, implement caching, and improve model efficiency.
4	Security	Basic authentication and limited data protection	Implement role-based access, data encryption, secure APIs, and regular security monitoring

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration of additional HDI-related datasets, improved visualizations, interactive charts, and exportable reports (PDF/CSV).	Jan – Jun 2027	Better insights, enhanced user experience, and improved analysis capabilities.
Phase 3	Implementation of advanced Machine Learning algorithms such as Random Forest and XGBoost, along with comparative country analysis dashboards	Jul – Dec 2027	Improved prediction accuracy and advanced analytical capabilities.
Phase 4	Mobile-friendly/PWA version, multilingual support, geospatial visualizations, and integration with external development data sources.	Jan – Jun 2028	Wider accessibility and richer insights , and increased scalability.